

Predicting Overall Indoor Environmental Quality Satisfaction in Classrooms Using Sensor, Occupant, and Contextual Data

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Abstract

This paper investigates whether overall indoor environmental quality (IEQ) satisfaction in classrooms can be predicted from environmental, contextual, occupant, and building-related variables. The analysis uses an open Belgian classroom dataset containing 6,834 repeated satisfaction assessments from 321 occupants across primary, secondary, and university classrooms. The original five-point overall IEQ scale is collapsed into three classes: dissatisfied, neutral, and satisfied. Five model families are compared under matched five-fold cross-validation, using macro F1 as the main selection metric because the target distribution is imbalanced. The best single model is an Extra Trees classifier, reaching macro F1 = 0.592, accuracy = 0.723, balanced accuracy = 0.594, and ordinal MAE = 0.314.

The confusion matrices show that satisfied votes are predicted most reliably, while dissatisfied and neutral responses remain harder to separate. Feature-importance and sensitivity analyses suggest that overall IEQ satisfaction is shaped by a combination of environmental, occupant, and contextual factors rather than by one dominant domain alone. Follow-up experiments with complete-case filtering, feature engineering, synthetic class balancing, and primary-school-specific training produce only modest gains and reveal domain shift between educational settings. The results quantify the performance and limits of classroom IEQ prediction and show how such models could support occupant-centered and energy-aware decision support in school buildings.

Keywords: indoor environmental quality; classroom comfort; occupant satisfaction; machine learning; Extra Trees; transfer learning; Belgian classroom dataset

1 Introduction

Indoor environmental quality (IEQ) describes the combined indoor conditions experienced by occupants, especially thermal comfort, indoor air quality, lighting, and acoustics. In classrooms, these conditions can affect students' comfort, engagement, well-being, and learning [1, 2, 3, 4, 5]. These effects are large enough to matter in everyday classroom operation, and they are not limited to obviously uncomfortable rooms. In a field study of 140 fifth-grade classrooms, each 1 L/s per person increase in ventilation within 0.9–7.1 L/s per person was associated with an increase of up to about 0.5% in mathematics scores. Within the ordinary 20–25 °C classroom-temperature range, each 1 °C decrease was associated with a further increase of about 0.5–0.6%. Published school-specific relationships also suggest larger effects under poorer conditions: lowering classroom temperature from 30 °C to 20 °C can increase performance by about 20%, and reducing classroom CO₂ from 2,100 ppm to 900 ppm can improve task speed by about 12% [6, 7, 8]. Recent thermal-comfort research reflects this need: PMV and adaptive comfort models are increasingly complemented by personal comfort models that use IoT data and machine learning to predict individual thermal preference [9, 10, 11]. These models are also moving into practice: personal-comfort-based control has been tested in occupied buildings, and one commercial system has used machine learning to translate warm/cool feedback into HVAC commands and thermostat schedules [12, 13]. However, this progress still focuses mainly on thermal comfort and HVAC control, not on overall classroom IEQ across air quality, lighting, and acoustics. In practice, classrooms are often monitored through separate indicators such as temperature, CO₂, illuminance, and sound level. These separate limits do not show whether students perceive the room as acceptable overall. They also miss operational trade-offs: shading can reduce overheating but lower daylight, and more ventilation can reduce CO₂ but affect thermal comfort, noise, and energy use [14, 15, 16, 17]. This paper therefore focuses on developing a model for predicting overall IEQ satisfaction: how students judge the

classroom environment as a whole, rather than as isolated responses to individual IEQ domains. The aim is to support better handling of energy-comfort trade-offs in classroom operation [18, 19, 20].

Several gaps in the IEQ literature motivate this focus. Reviews show that overall IEQ satisfaction is affected by several domains at the same time, but many existing models do not fully capture these interactions and have only limited validation across real buildings [15, 16, 20, 17]. Many models also rely on subjective votes for separate domains, such as thermal, air-quality, visual, or acoustic satisfaction. These inputs are useful for research, but they are not always available in routine classroom monitoring, where decisions must often rely on sensor and context data. Recent sensor-based studies show that monitored environmental data can predict perceived IEQ, including student perceptions in schools [21, 22, 5]. However, classrooms remain less developed than offices or commercial buildings: students have limited control over the indoor environment, may respond differently from adults, and existing studies are difficult to compare or reuse [23, 24, 25, 26]. The open Belgian classroom dataset helps address this gap by linking repeated overall and domain satisfaction votes with environmental, personal, and contextual variables across primary, secondary, and university classrooms [27]. This paper uses that dataset to test how well overall IEQ satisfaction can be predicted from classroom sensor, occupant, and context data, whether the prediction can be personalized, and how well models generalize across educational settings.

1.1 Research Questions

1. **RQ1: What prediction accuracy can be achieved for IEQ satisfaction using sensor and contextual data?**
2. **RQ2: Which sensor, occupant, and contextual variables are most influential to IEQ satisfaction?**
3. **RQ3: How many occupant votes are required to achieve effective personalization for individual occupants or specific classrooms?**
4. **RQ4: What performance loss occurs when prediction models are transferred across educational settings?**

1.2 Contribution

The practical value of this work is to move classroom IEQ assessment from isolated threshold checking toward occupant-centered and energy-aware decision support. Instead of evaluating temperature, CO₂, light, or sound separately, the study asks whether available classroom data can indicate when students are likely to perceive the indoor environment negatively as a whole.

- It provides a first step toward future energy-comfort optimization by predicting overall IEQ satisfaction instead of treating temperature, CO₂, light, and sound as separate thresholds.
- It evaluates how well a three-class model can predict overall IEQ satisfaction and identifies where the prediction remains uncertain (RQ1).
- It identifies which measured variables are most useful for dashboard warnings and intervention prioritization (RQ2).
- It tests whether prediction can be improved by adapting the pooled model to a specific classroom or occupant (RQ3).
- It evaluates cross-setting generalization by testing whether models trained in one educational setting remain reliable in another; if an independent classroom dataset becomes available, this step will be extended to external transfer-learning validation (RQ4).

2 Dataset Description

The source material is the open Belgian classroom dataset with 6,834 satisfaction assessments from 321 occupants across 11 classrooms in three educational settings: primary school, secondary school, and a university test lecture

room [27]. It was collected in Flanders, Belgium: in three secondary-school classrooms in Haacht from April to June 2021, seven primary-school classrooms in five schools from October to December 2021, and a test lecture room at KU Leuven Ghent Campus from November 2021 to May 2022 [27]. The secondary-school campaign covered intermediate-season conditions, the primary-school campaign covered heating-season conditions, and the test lecture room covered heating and intermediate-season conditions [27].

Occupant feedback was collected through repeated right-here-right-now online surveys. The surveys recorded five-point satisfaction votes for thermal, indoor-air-quality, acoustic, visual, and overall IEQ conditions, together with personal and contextual information such as occupant ID, age, gender, clothing, seating position, survey moment, activity, and classroom or group; each vote was linked to the nearest environmental measurements in time [27].

The environmental monitoring setup differed between the three campaigns. In the secondary school, three Renson Sense IEQ devices were placed in each classroom and measured air temperature, relative humidity, CO₂, light intensity, and sound level at 5-minute intervals; outdoor temperature was measured separately [27]. In the primary schools, classroom IEQ was monitored using HOBO-MX, HOBO U12-O12, and Netatmo smart weather station devices, also at 5-minute intervals, with an additional CO₂ monitoring device in each classroom to inform pupils and teachers about IAQ conditions [27]. In the test lecture room, building sensors and a weather station measured indoor and outdoor environmental conditions at 1-minute intervals, an Ethern NEMO IAQ sensor measured IAQ variables at 10-minute intervals, and a Testo 480 thermal comfort meter measured additional thermal variables such as air velocity and mean radiant temperature [27].

This study uses the combined simplified version of the dataset, which keeps variables available across the three cases. It includes the original five-point overall IEQ satisfaction vote and a common set of environmental, occupant, and contextual variables, such as air temperature, relative humidity, CO₂, illuminance, LAeq, clothing, age, gender, seating position, classroom type, and survey timing.

Supervisor note: The following graphs are here only to help understanding the dataset.

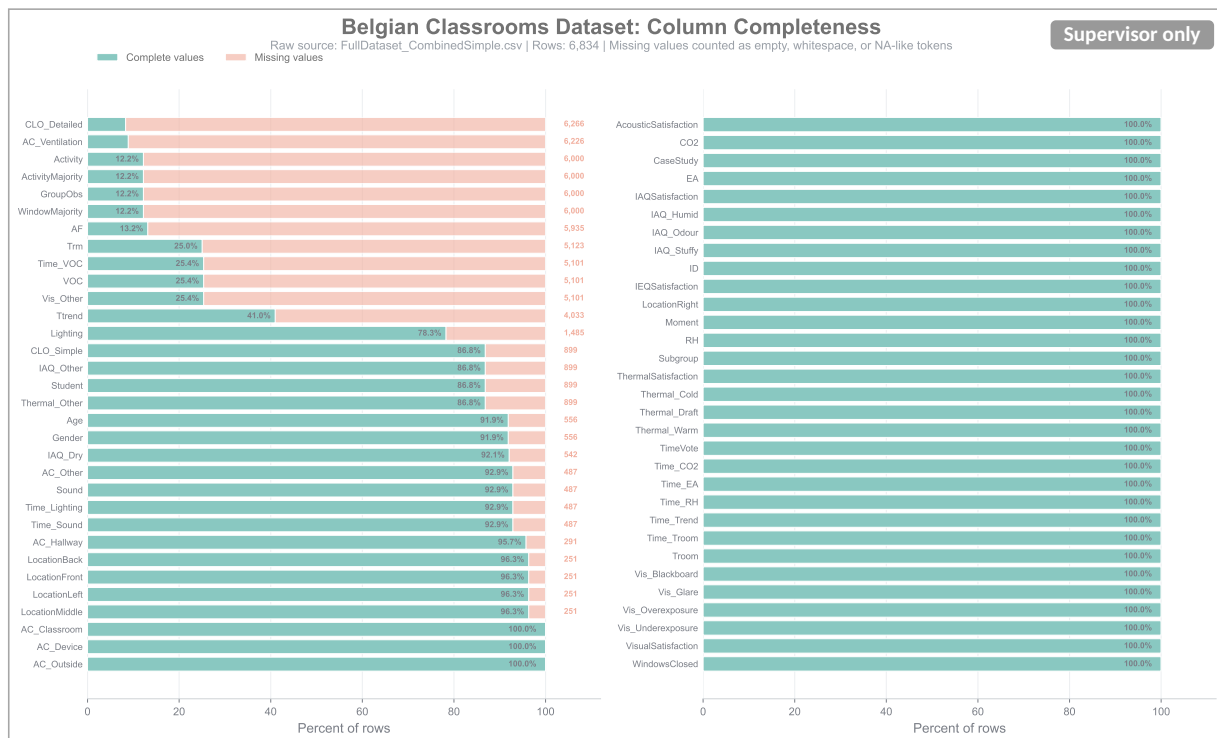


Figure 1: Supervisor-only supporting figure: raw dataset column completeness across the shared classroom dataset.

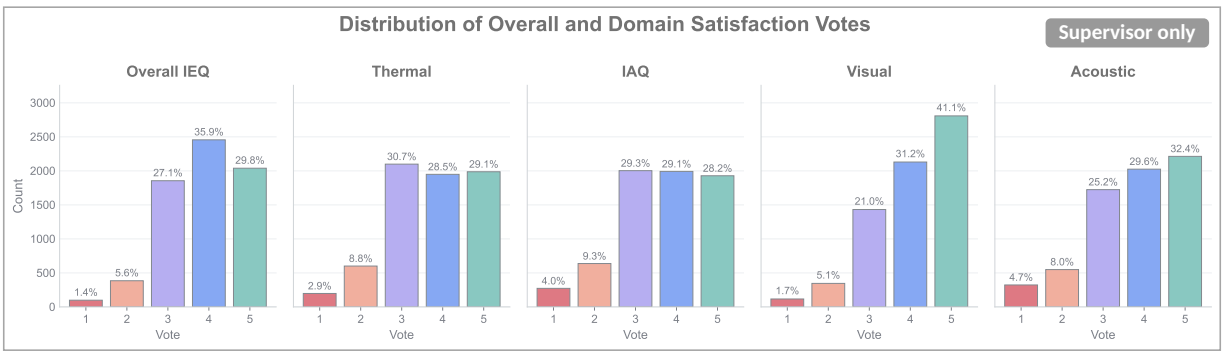


Figure 2: Supervisor-only supporting figure: satisfaction vote distributions across the overall IEQ target and the four domain targets on the original five-class scale.

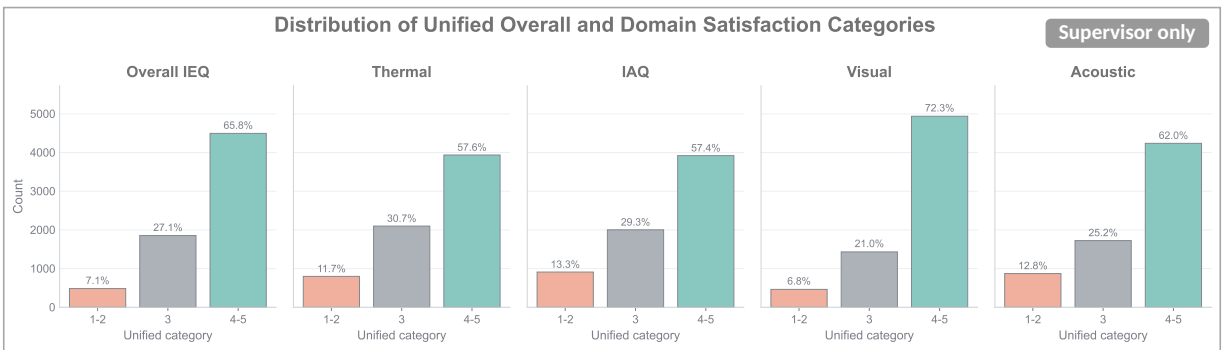


Figure 3: Supervisor-only supporting figure: unified satisfaction-category distributions across the shared target set.

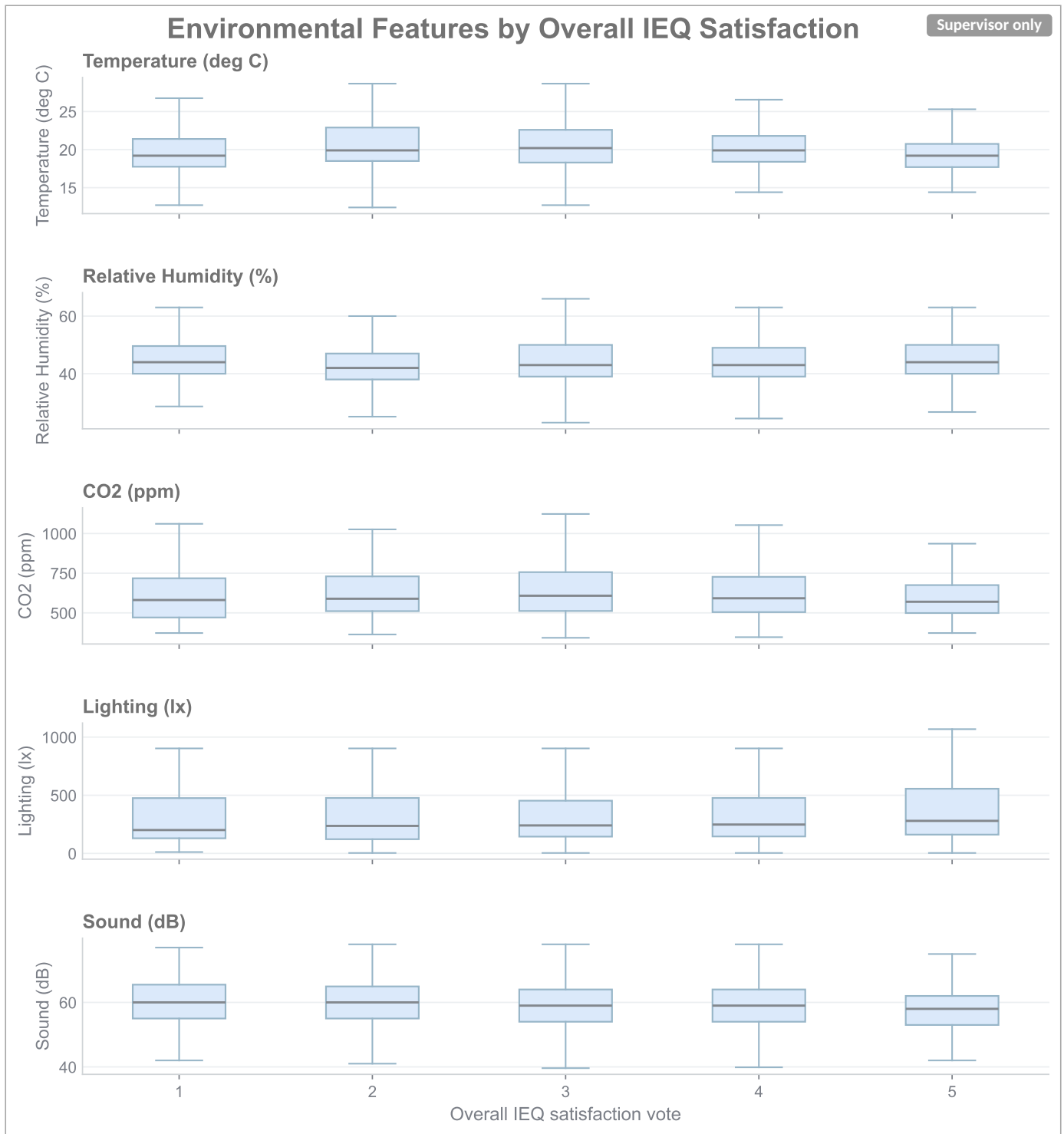


Figure 4: Supervisor-only supporting figure: environmental feature boxplots for the IEQ paper variables.

3 Methodology

3.1 Prediction Task and Model Inputs

Before imputation, the model input and output variables were defined explicitly. The output variable was the overall IEQ satisfaction vote. The input variables were grouped into sensor-based measurements (Temperature, RH, CO2, Lighting, Sound, and Ttrend), occupant descriptors (Age, Gender, Student, and CL0), seating context (classroom lo-

cation), and other contextual variables (EA, Vote hour, Vote weekday, Moment, and AC on/off). In the model-ready dataset, classroom location was represented by five binary columns: LocationBack, LocationFront, LocationLeft, LocationRight, and LocationMiddle. Almost all variables that were consistently available in the combined simplified dataset were retained. The remaining variables were excluded mainly because they were not available across the three cases, had too many missing values for a comparable model, or would introduce leakage. In particular, domain-specific satisfaction votes were not used as predictors because they are subjective responses close to the target variable, while trace and grouping identifiers, including Occupant ID, Group, and Subgroup, were excluded to avoid identity- or classroom-specific leakage and unintended personalization to individual occupants or classrooms. The completeness of the retained inputs is summarized in Figure 5.

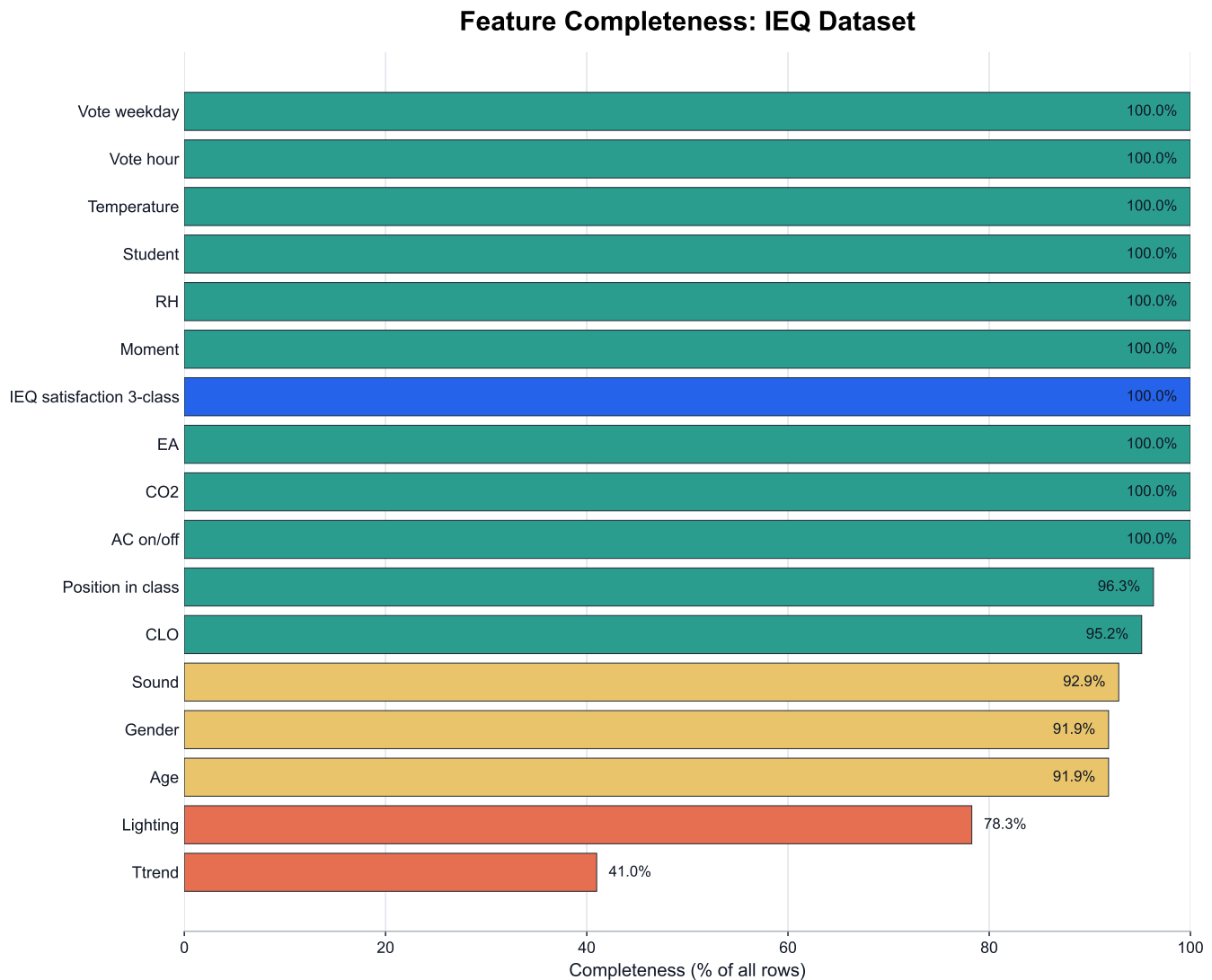


Figure 5: Feature completeness for the variables currently retained in the IEQ paper.

3.2 Data Cleaning and Imputation

Missing values were filled using fixed, reproducible rules. Missing classroom position was set to Middle. Clothing insulation (CLO) was imputed by the median within each Group, with the global dataset median used as fallback. In the survey and dataset documentation, clothing is recorded as either a simple three-level category (light, neutral, warm) or a more detailed clothing description, not as a directly measured insulation value. Here, the simple categories were mapped to approximate clo values of 0.44, 0.64, and 0.81 based on standard clothing-insulation tables; these values should therefore be interpreted as representative approximations rather than observed measurements [27, 28, 29].

Temperature trend, sound level, and lighting were imputed by the median within Group + Subgroup, then by Group, and finally by the global median as fallback. Age was first inferred from other records with the same Occupant ID using that occupant’s median reported age; if no age was available for that ID, the fallback sequence was the median within Group + student-status, then Group, then the global median. Gender was first copied from other records with the same Occupant ID; if no gender was available for that ID, it was assigned by a seeded random draw from the gender distribution within the same Group, with the overall dataset distribution used as fallback. The original five-category overall IEQ scale (1 = very dissatisfied, 2 = dissatisfied, 3 = ok, 4 = satisfied, 5 = very satisfied) was collapsed into three classes for modeling purposes, with categories 1 and 2 merged as dissatisfied, category 3 retained as neutral, and categories 4 and 5 merged as satisfied. The distributions on both scales are shown in Figure 6.

Overall IEQ Satisfaction Vote Distribution

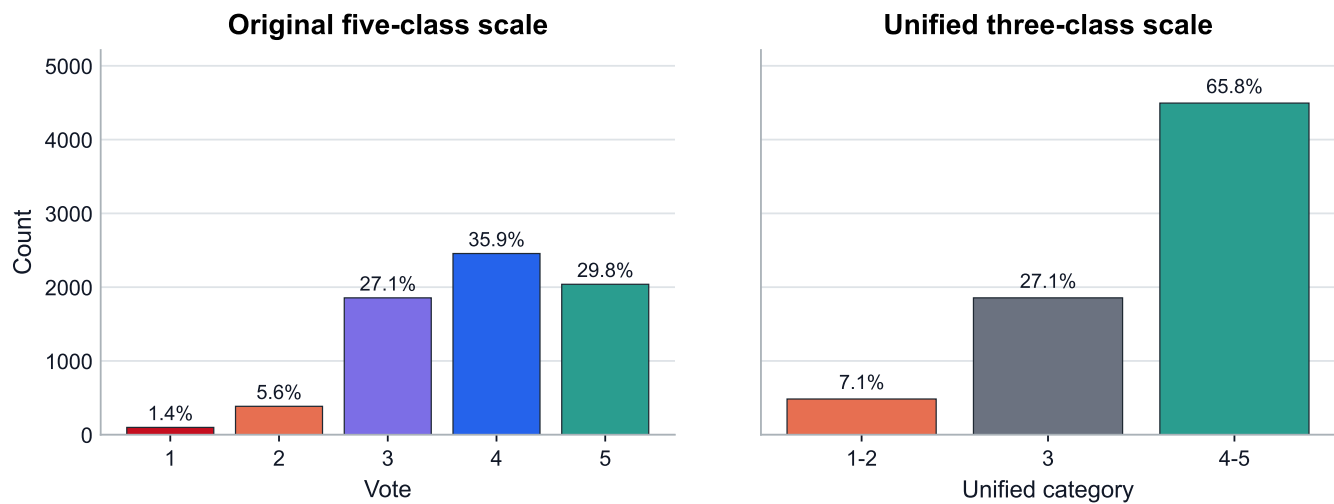


Figure 6: Original and unified overall IEQ satisfaction vote distributions in the current dataset.

3.3 Model Validation Strategy

Previous IEQ reviews show that differences in measurement protocols, building contexts, sample sizes, and modeling approaches make direct comparison between studies difficult [15, 16, 20]. To ensure a fair comparison, the same validation protocol was applied to all model families.

Model performance was evaluated using five-fold cross-validation on the cleaned dataset with the defined predictor set and three-class IEQ satisfaction target. The folds were kept identical for all model families and tuning stages. Preprocessing was fitted within each training fold to prevent information leakage. Because the three target classes were unevenly distributed, macro F1 was used as the primary selection metric: it averages class-level F1 scores without weighting them by class frequency, allowing minority classes to contribute equally to model evaluation [30]. Accuracy is reported only as a secondary metric because it can be inflated by the dominant satisfied class. Balanced accuracy provides a complementary class-imbalance diagnostic. Ordinal mean absolute error (MAE) was also reported by coding the ordered classes as dissatisfied = 0, neutral = 1, and satisfied = 2, so larger class-distance errors were penalized more strongly than adjacent-class errors.

Macro F1 will also be used as the main metric for personalization and transfer analysis. Personalization will compare the pooled model with classroom- and occupant-specific variants and estimate how many feedback votes are needed for useful adaptation. Transfer learning will first evaluate the Belgian classroom model on a new classroom or school from an independent dataset. It will then test whether a limited number of local feedback votes improves performance in the target setting.

3.4 Benchmark Model Training

The benchmark compared five supervised model families for tabular classification: Logistic Regression, Gradient Boosting, ANN/MLP, Random Forest, and Extra Trees. All five model families were evaluated on the same predictor set, cross-validation folds, and primary selection metric. Hyperparameters were tuned for each model family before the final benchmark comparison.

3.5 Hyperparameter Tuning and Final Model Optimization

Hyperparameter tuning was treated as part of the modeling protocol rather than as a result. The workflow proceeded in three steps:

1. Tune all five model families using the same cross-validation protocol and macro F1 objective.
2. Select the strongest tuned model family as the main reference model.
3. Test additional optimization strategies for the selected model.

After the strongest tuned model was selected, additional optimization strategies were tested for that model. These included manual temperature and CO₂ threshold features, ensemble and ordinal alternatives, complete-case row filtering, and synthetic class balancing. The selected Extra Trees model was also interpreted using complementary feature-importance approaches, including model-based importance, permutation importance, and SHAP-based importance grouped back to the original input variables. These analyses were used to identify potentially useful predictors for monitoring and decision support, not to claim causal effects.

3.6 Follow-Up Analyses

3.6.1 Personalization Analysis

Personalization will be evaluated for new classrooms and new occupants. For each target classroom or occupant, all of its records will be excluded from the base model training. A limited number of target feedback votes will then be used for adaptation, and the remaining target votes will be used for testing. Learning-curve experiments will estimate how many feedback votes are needed before personalization becomes useful. Identifiers such as `Occupant ID`, `Group`, and `Subgroup` will be used only to define personalization groups and evaluation splits, not as predictors in the general model.

3.6.2 Transfer-Learning and Cross-Setting Validation

Transfer learning will be evaluated using the Norwegian classroom dataset used by Alam et al. [5]. The model will first be trained on the full Belgian classroom dataset. Before evaluation, the variables and target classes will be harmonized across the Belgian and Norwegian datasets. The Belgian-trained model will then be tested on the Norwegian data before and after adaptation with a limited number of local feedback votes.

4 Results

4.1 Benchmark Model Comparison

Table 1 summarizes the performance after hyperparameter tuning. Extra Trees achieved the highest macro F1, followed closely by Random Forest, while the MLP, Gradient Boosting, and Logistic Regression models performed less well on the class-balanced selection metric. The selected hyperparameter configurations are reported in Table 2 for reproducibility.

Table 1: Five-fold cross-validation summary for the current all-school IEQ benchmark after hyperparameter tuning.

Model	Macro F1	Accuracy	Bal. acc.	Ordinal MAE
Extra Trees	0.592	0.723	0.594	0.314
Random Forest	0.587	0.714	0.590	0.326
ANN / MLP	0.494	0.691	0.476	0.352
Gradient Boosting	0.497	0.695	0.473	0.347
Logistic Regression	0.402	0.483	0.468	0.669

Table 2: Main selected hyperparameter settings for the IEQ model families.

Model	Main selected settings
Extra Trees	500 trees, entropy criterion, bootstrap sampling, no maximum depth, full feature sampling, <code>max_samples=0.9</code> , balanced class weights
Random Forest	300 trees, Gini criterion, no maximum depth, square-root feature sampling, <code>min_samples_leaf=4</code> , <code>min_samples_split=10</code> , balanced subsample class weights
ANN / MLP	Hidden layers with 64 and 32 neurons, tanh activation, <code>alpha=0.001</code> , batch size 64, learning rate 0.003
Gradient Boosting	220 estimators, learning rate 0.08, max depth 3, <code>min_samples_leaf=10</code> , subsample 0.85, full feature sampling
Logistic Regression	<code>C=0.03</code> , L-BFGS solver, balanced class weights

The full scikit-learn configurations are available in the accompanying code and output tables.

Extra Trees was therefore selected as the main reference model for the following analyses. Its advantage over Random Forest was small but consistent for the primary macro F1 metric and the secondary accuracy and ordinal MAE metrics.

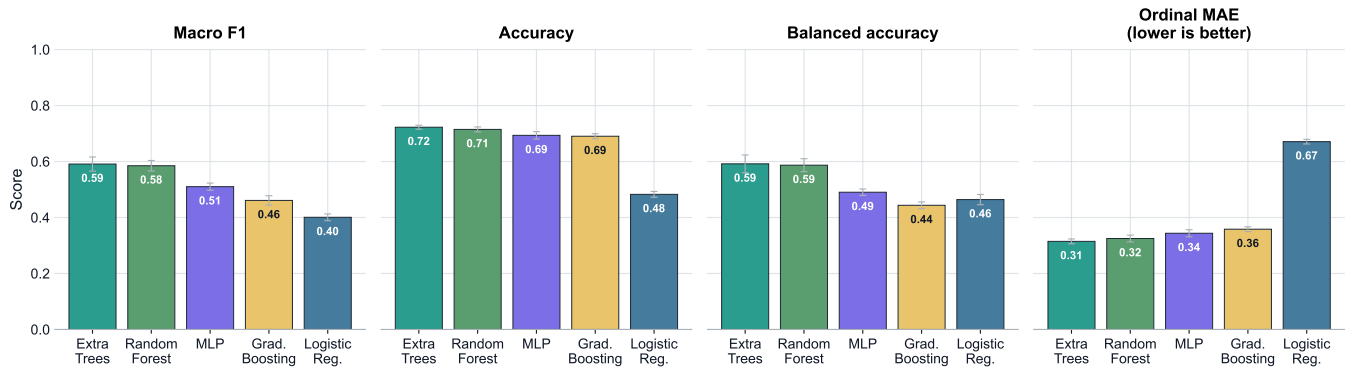


Figure 7: Performance comparison for the IEQ benchmark models after hyperparameter tuning.

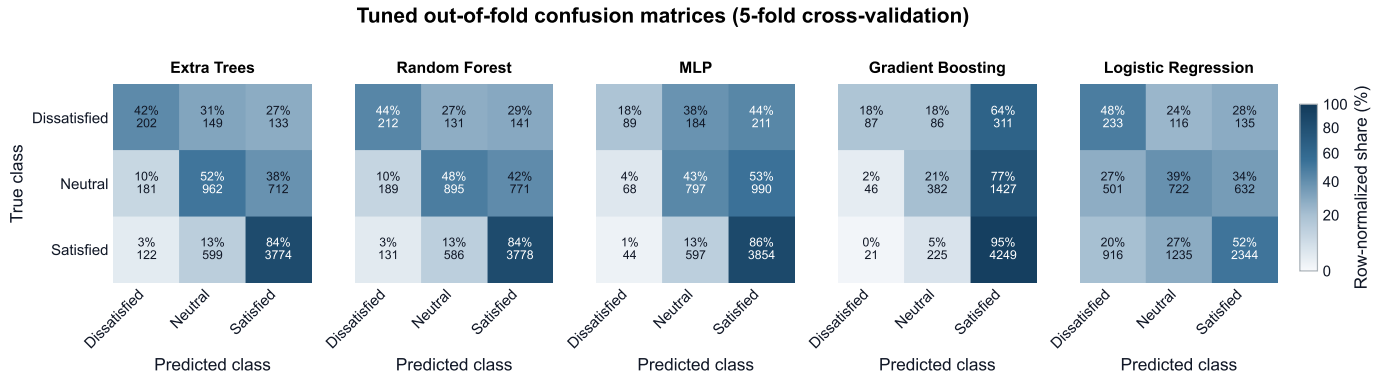


Figure 8: Aggregated confusion matrices for the tuned IEQ benchmark models.

The confusion matrices show that satisfied votes are predicted most reliably, while dissatisfied and neutral votes remain harder to detect. Extra Trees gives the most balanced class-level performance, correctly detecting 42.1% of dissatisfied votes and 53.0% of neutral votes. Gradient Boosting and ANN/MLP assign too many observations to the satisfied class, which reduces minority-class performance while maintaining relatively high accuracy. This supports the use of macro F1 instead of accuracy alone.

4.2 Extra Trees Fine Tuning Results

Several optimization experiments were compared with the selected all-school Extra Trees model:

Manual temperature and CO₂ feature engineering. We added three threshold indicators: CO₂ > 1000 ppm, $T < 18^{\circ}\text{C}$, and $T > 24^{\circ}\text{C}$. Results: macro F1 = -0.006, accuracy = -0.005, balanced accuracy = -0.006, and ordinal MAE = +0.006. The indicators therefore did not improve the current baseline.

Imputation-indicator columns. We added imputation flags for incomplete predictors plus a row-level imputation count. Results: macro F1 = +0.012, accuracy = +0.007, balanced accuracy = +0.012, and ordinal MAE = -0.007. This was the strongest simple improvement.

Soft-voting ensemble. We combined Extra Trees and Random Forest probabilities; the best weight used 80% Extra Trees and 20% Random Forest. Results: macro F1 = +0.002, accuracy = +0.001, balanced accuracy = +0.001, and ordinal MAE = +0.000. The gain was negligible.

Ordinal classification experiment. We tested two-threshold ordinal decompositions for $y > \text{dissatisfied}$ and $y > \text{neutral}$. Extra Trees results: macro F1 = -0.006, accuracy = +0.007, balanced accuracy = -0.008, and ordinal MAE = -0.003. This stayed supplementary because macro F1 did not improve.

Data-quality and row-filtering strategies. We tested complete-case filters for originally observed sensor values. The main-sensor filter kept 5,349 rows. Results: macro F1 = +0.014, accuracy = -0.002, balanced accuracy = +0.015, and ordinal MAE = +0.002. The stricter Lighting + Sound + Ttrend filter kept 2,314 rows. Results: macro F1 = +0.009, accuracy = -0.021, balanced accuracy = +0.009, and ordinal MAE = +0.018. We did not keep these filters because accuracy did not improve and other indicators improved only slightly.

Synthetic class balancing. We tried to reduce the class imbalance by adding extra copies of under-represented training samples. This was done only inside each training fold, so the validation data remained unchanged. The best setting raised the smaller classes to 50% of the majority class size. It slightly improved macro F1 +0.003 and balanced accuracy +0.015, but reduced accuracy -0.006 and worsened ordinal MAE +0.013, so it was not retained.

Class-weight and decision-threshold tuning. We screened class weights and probability multipliers. The best class-weight setting, {dissatisfied: 3, neutral: 2, satisfied: 1}, gave: macro F1 = +0.005, accuracy = +0.013, balanced accuracy = -0.016, and ordinal MAE = -0.017. Nested threshold tuning gave: macro F1 = +0.002, accuracy = -0.001, balanced accuracy = -0.009, and ordinal MAE = -0.004. The effect was small and mixed across metrics.

Ten-fold validation check. Five-fold cross-validation was used for model screening. After the final workflow was fixed, the selected Extra Trees model was re-evaluated with ten stratified folds. The ten-fold baseline reached macro F1 = 0.596, accuracy = 0.724, balanced accuracy = 0.598, and ordinal MAE = 0.313. With imputation indicators, performance changed by macro F1 = +0.012, accuracy = +0.008, balanced accuracy = +0.015, and ordinal MAE = -0.010. The ten-fold setup was used only for the final performance estimate, not for model selection.

Table 3: Extra Trees optimization summary.

Experiment	Macro F1	Accuracy	Bal. acc.	Ordinal MAE
Main all-school Extra Trees baseline	0.592	0.723	0.594	0.314
Threshold T/CO ₂ feature engineering	-0.006	-0.005	-0.006	+0.006
Imputation-indicator columns	+0.012	+0.007	+0.012	-0.007
Soft-vote ET 0.80 + RF 0.20	+0.002	+0.001	+0.001	0.000
Ordinal Extra Trees decomposition	-0.006	+0.007	-0.008	-0.003
Ordinal Random Forest decomposition	-0.044	-0.011	-0.040	+0.023
Best class-weight Extra Trees	+0.005	+0.013	-0.016	-0.017
Best class-weight + nested threshold	+0.002	-0.001	-0.009	-0.004
Main-sensor complete-case filter	+0.014	-0.002	+0.015	+0.002
Lighting + Sound + Ttrend complete-case filter	+0.009	-0.021	+0.009	+0.018
Best synthetic sampling screen	+0.003	-0.006	+0.015	+0.013
Ten-fold validation check	+0.003	+0.001	+0.004	-0.001

In Table 3, the baseline row reports the original metric values. The other rows report changes relative to this baseline. Black rows show retained modelling steps, while grey rows show supplementary checks. The final workflow kept imputation indicators, class-weight tuning, and nested threshold adjustment. Ten-fold cross-validation was used only for the final performance estimate. For ordinal MAE, negative changes indicate improvement.

4.3 Final Extra Trees Model Results

The final Extra Trees model includes imputation-indicator predictors, class-weight adjustment, and nested threshold tuning. After these choices were fixed, performance was estimated using ten-fold stratified validation. Threshold multipliers were selected only inside the training part of each fold.

Table 4 reports the final cross-validated performance. The model reached macro F1 = 0.623, accuracy = 0.740, balanced accuracy = 0.625, and ordinal MAE = 0.291.

Table 4: Final Extra Trees model performance under ten-fold stratified cross-validation.

Model	Macro F1	Accuracy	Bal. acc.	Ordinal MAE
Final Extra Trees	0.623	0.740	0.625	0.291

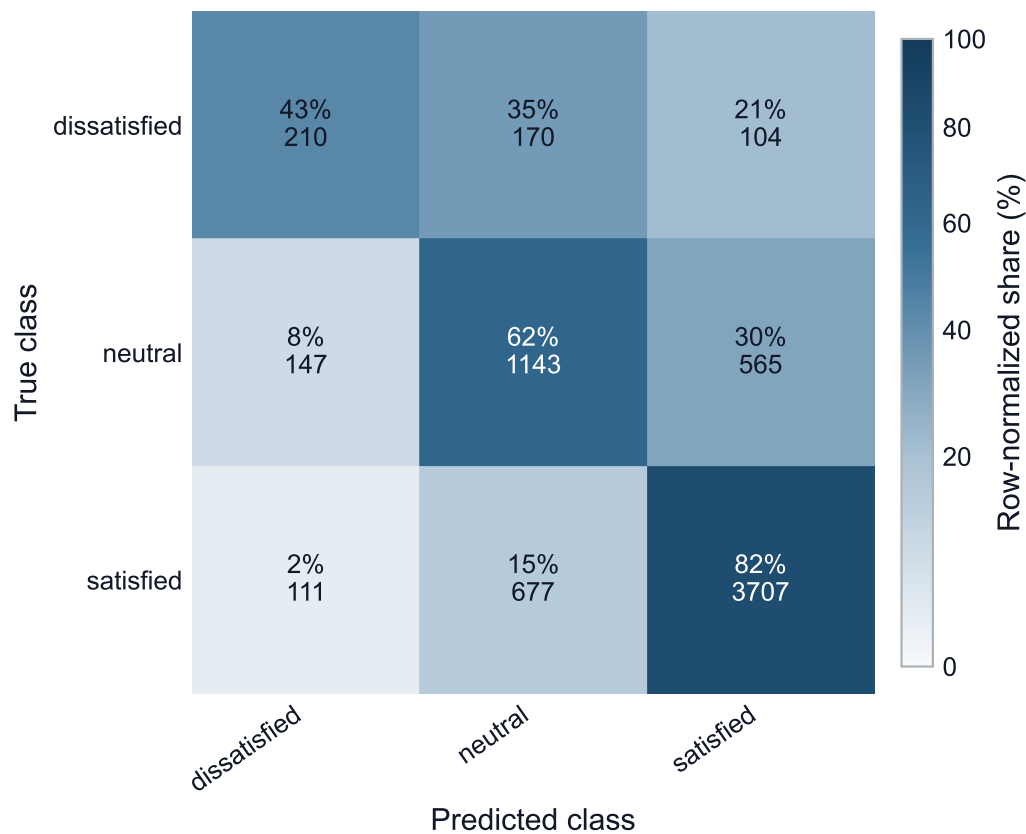


Figure 9: Out-of-fold confusion matrix for the final Extra Trees model. Cells show row percentages and counts.

Figure 9 shows that satisfied votes remain the easiest class. The model correctly identified 3,687 of 4,495 satisfied votes, giving satisfied recall of 0.820. Neutral recall was 0.613, and dissatisfied recall was 0.438. The main remaining errors were between neighboring classes: 568 neutral votes were predicted as satisfied, and 697 satisfied votes were predicted as neutral. This means the model is more reliable for broad IEQ screening than for exact separation of adjacent satisfaction states.

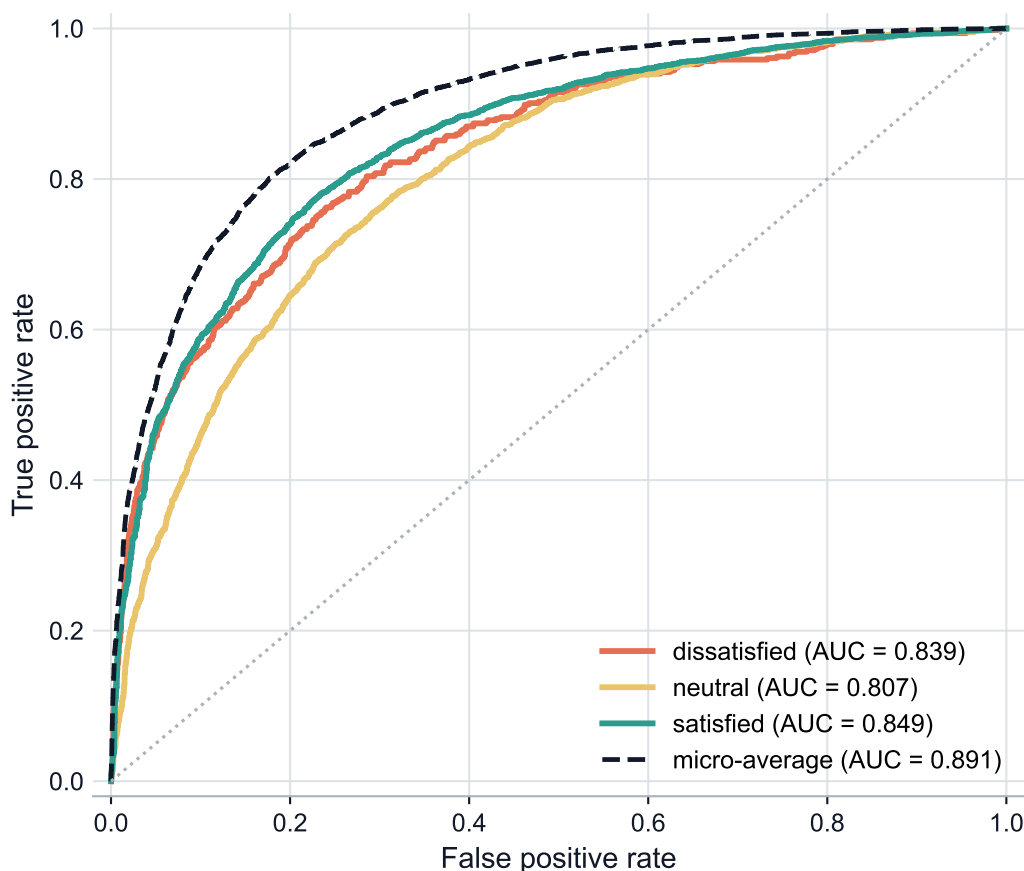


Figure 10: One-vs-rest ROC curves for the final Extra Trees model using ten-fold out-of-fold probabilities.

Figure 10 shows the ranking ability of the final model for each class. The one-vs-rest AUC values were 0.839 for dissatisfied, 0.807 for neutral, and 0.849 for satisfied votes. The micro-average AUC was 0.891.

After cross-validation, the same final workflow was fitted once more on the full dataset of 6,834 observations. This full-data model is the version saved for later prediction or deployment. Its settings use the probability multipliers selected from the cross-validation results: {dissatisfied: 1.5, neutral: 1.2, satisfied: 1.0}. No additional performance score is reported for this fitted model, because it is trained on all available observations and therefore has no separate validation fold.

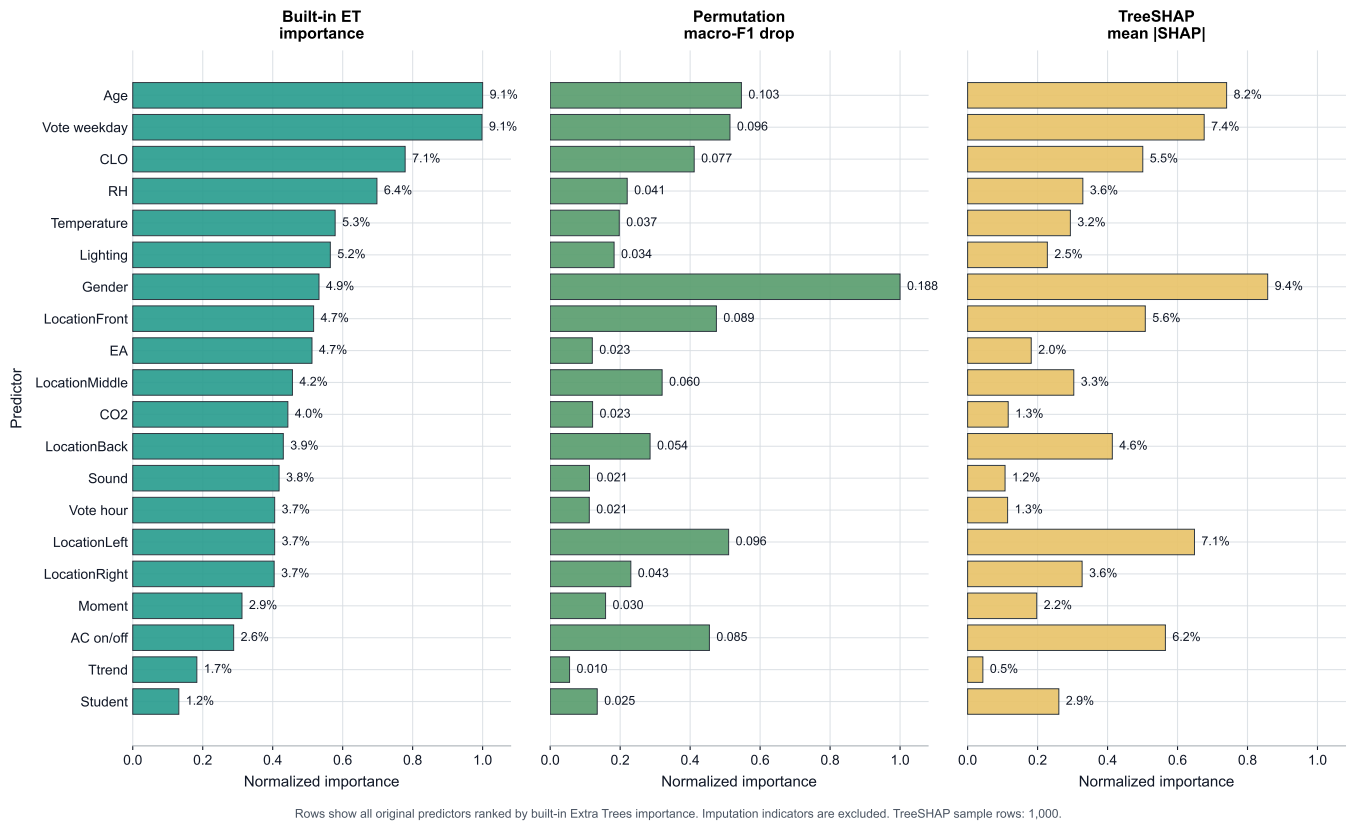


Figure 11: Feature-importance comparison for the final Extra Trees IEQ prediction model. All original predictors are shown; imputation indicators are excluded.

Figure 11 compares built-in Extra Trees importance, permutation importance, and TreeSHAP importance. The rows are sorted by built-in Extra Trees importance. The highest built-in values were found for Vote weekday, Age, CLO, relative humidity, temperature, lighting, and Gender. Some predictors, especially Gender, LocationLeft, and AC on/off, had stronger permutation or TreeSHAP importance than built-in importance. This shows that the ranking depends on the interpretation method. The figure should therefore be read as model interpretation, not as causal evidence.

We have to decide which method to public. For our results Built-in importance looks the best, because RH, Temperature, Lighting, CO2 and sound score quite high, which would show that the satisfaction is predicted more on sensor data than in the other two methods.

4.4 Personalization Results

These personalization experiments are exploratory. They were added to check whether the final Extra Trees modelling strategy can adapt to a new occupant or classroom after a small number of target-specific feedback votes. The design was target-held-out: all votes from the target occupant or classroom were removed from the zero-shot training set, the first k chronological target votes were used for adaptation, and later target votes were used for testing.

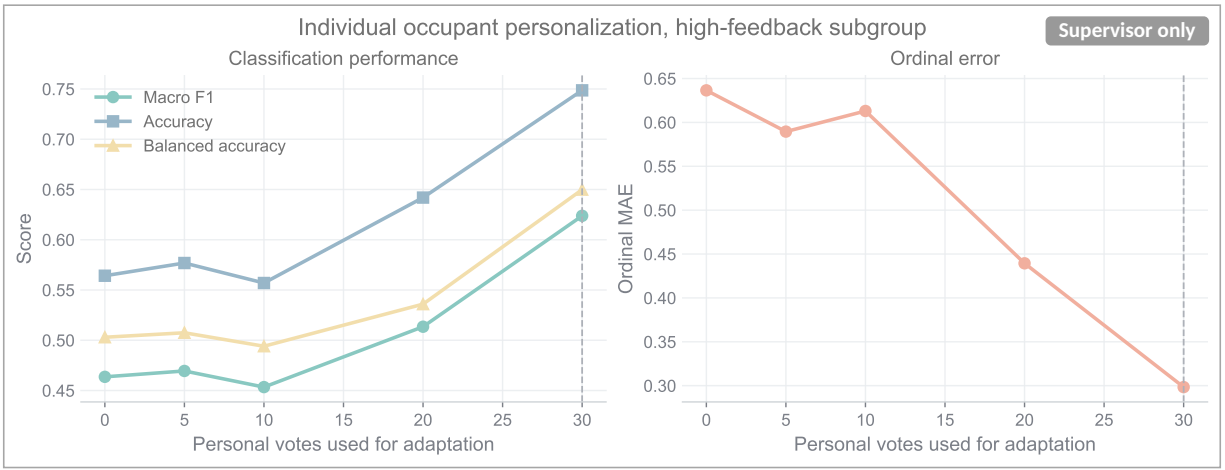


Figure 12: Supervisor-only exploratory figure: individual occupant personalization learning curve for the model with classroom-position predictors.

For individual occupants, the curve improves mainly after larger feedback windows. This could mean that target feedback helps the model adapt, but it could also mean that the original model already contains indirect occupant information. Even though direct identifiers were excluded, variables such as age, student status, classroom position, and repeated response patterns may partly identify some pupils.

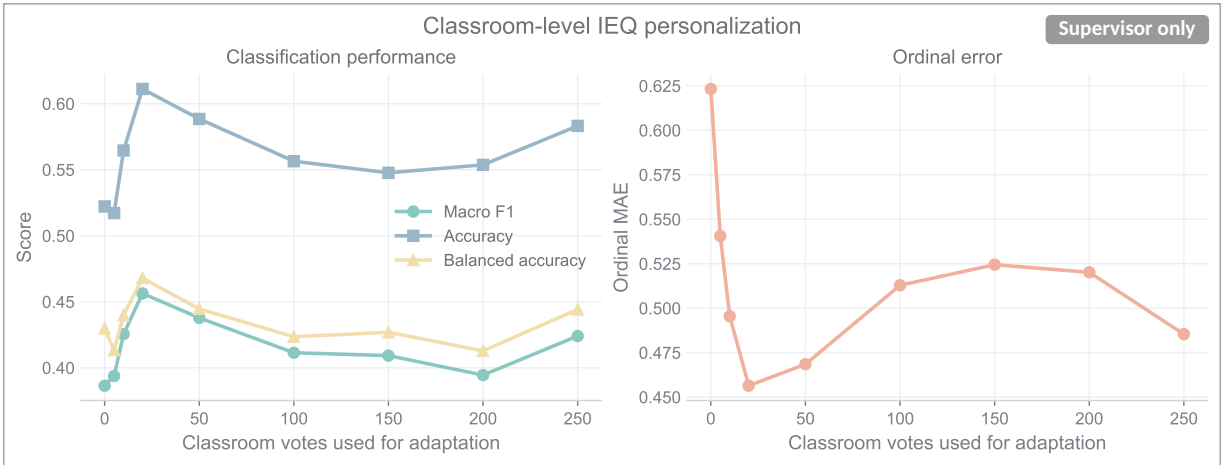


Figure 13: Supervisor-only exploratory figure: classroom personalization learning curve for the model with classroom-position predictors.

The classroom-level curve is weaker and not monotonic. It improves most clearly at a small feedback window, but adding more classroom votes does not steadily improve the result. This makes the current weighted-refit personalization method useful as a diagnostic check, but not yet strong enough to present as a robust article-ready personalization result.

The main interpretability concern is classroom position. The original reason for using position was that pupils near windows or specific room zones can experience different sun exposure, glare, draught, ventilation, or local thermal conditions. However, the personalization results suggest that position may also help the model recognize recurring pupils or classroom patterns. For example, the model might partly learn that a 13-year-old pupil sitting on the left side of a specific classroom often answers “satisfied”, rather than learning a general environmental effect of sitting on the left. To check this, I trained an additional Extra Trees model without the five classroom-position predictors.

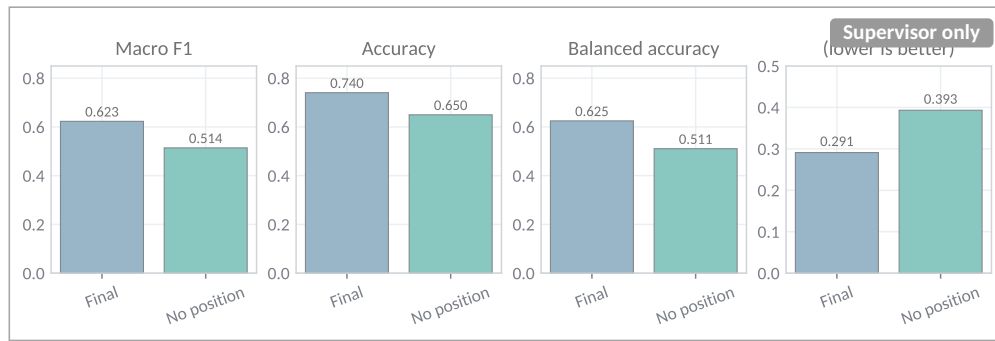


Figure 14: Supervisor-only exploratory figure: final Extra Trees performance with and without classroom-position predictors. Removing the five classroom-position columns reduced macro F1 from 0.623 to 0.514 and accuracy from 0.740 to 0.650.

This ablation shows that classroom position carries substantial predictive signal. It does not prove whether the signal is a real spatial IEQ effect or an indirect identity signal. I would therefore treat the personalization section with caution and be ready to explain this limitation if reviewers ask about leakage, repeated occupants, or indirect personalization.

4.5 Transfer-Learning and Cross-Setting Validation Results

The transfer-learning and cross-setting validation will be done after the Norwegian dataset is available. (They should send us the dataset until June 21st)

5 Preliminary Literature Benchmark

This section compares the final results with related IEQ prediction studies. The final Extra Trees model reached macro F1 = 0.623, accuracy = 0.740, balanced accuracy = 0.625, and ordinal MAE = 0.291 for the three-class overall IEQ target. These values are lower than some reported benchmarks, but the comparison is limited. This study predicts individual classroom votes across primary, secondary, and university classrooms, and it mainly uses sensor, occupant, and context variables rather than subjective votes from other IEQ domains.

Higher scores in the literature are often reported for easier or different settings, such as binary targets, aggregated dissatisfaction rates, person- or building-specific models, or models using subjective thermal, air-quality, visual, and acoustic satisfaction as inputs [19, 16]. In this context, the current performance appears realistic. The model is weaker for dissatisfied and neutral votes, but it still shows that sensor and context data can support broad IEQ satisfaction screening. Macro F1 and balanced accuracy are therefore important alongside accuracy, because they show the weaker performance on minority classes.

6 Submission Notes

The most suitable target journal currently seems to be *Building and Environment*, because the paper combines IEQ, occupant satisfaction, classroom measurements, and building-operation implications. Possible fallback journals are *Indoor and Built Environment*, *Journal of Building Engineering*, and *Energy and Buildings*. *Indoor and Built Environment* fits the IEQ and occupant-response focus, *Journal of Building Engineering* fits the applied building-performance and modelling angle, and *Energy and Buildings* could be suitable if the paper emphasizes the future use of the model for balancing satisfaction and energy use.

7 Discussion

8 Future Work

Future work should move the model closer to real building operation. A practical next step would be to test it in a classroom monitoring dashboard, where predicted satisfaction could be compared with new student feedback and used to identify problematic rooms, lessons, or operating conditions. Such testing would show whether the model is useful not only as a retrospective benchmark, but also as a simple decision-support tool for school operation.

Future studies should also compare ML-based IEQ approaches with more conservative building-operation strategies.

This would help determine whether IEQ satisfaction models can bring measurable energy benefits when used as part of more complex building control, and what benefits different satisfaction-energy trade-offs could provide. Further progress will require more data from different building types, schools, seasons, and climate zones. It would also be useful to define a more unified set of variables that should be collected in IEQ studies, so that future datasets are easier to combine and compare.

9 Conclusion

This study evaluated whether overall IEQ satisfaction in classrooms can be predicted from sensor, occupant, and contextual data. Using 6,834 repeated votes from the Belgian classroom dataset, the original five-point overall IEQ satisfaction scale was transformed into a three-class prediction task. Among the tested model families, Extra Trees gave the strongest and most stable performance. After the retained optimization steps, the final model reached macro F1 = 0.623, accuracy = 0.740, balanced accuracy = 0.625, and ordinal MAE = 0.291 under ten-fold cross-validation.

The results show that overall classroom IEQ satisfaction can be predicted with useful but still limited accuracy from routinely collectable variables. Satisfied votes were identified most reliably, while dissatisfied and neutral votes remained more difficult to separate. This indicates that the model is more suitable for broad screening and dashboard-based warning support than for exact classification of every individual response. The feature-importance analyses also suggest that overall IEQ satisfaction is not driven by a single environmental variable, but by a combination of environmental conditions, occupant characteristics, and contextual factors.

The study also shows the limits of the current approach. Simple feature engineering, synthetic class balancing, ordinal decomposition, and ensemble variants produced only modest or inconsistent improvements. Exploratory personalization checks further suggested that adapting the model to specific occupants or classrooms is harder for overall IEQ satisfaction than expected from thermal-comfort personalization studies. External transfer validation remains necessary before the model can be considered robust across schools, seasons, and measurement protocols.

Overall, the findings support the use of machine learning as a complementary tool for occupant-centered classroom operation. The model should not replace direct feedback or established environmental measurements, but it can help combine multiple sensor and context signals into a single estimate of likely overall IEQ satisfaction. This makes it a useful first step toward classroom monitoring systems that consider both comfort and energy-related operational decisions.

10 References

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